Rebecca Woody

Ph.D. Candidate, Astronomy & Astrophysics

Cambridge, MA | rebecca.woody@cfa.harvard.edu | rebeccawoody.github.io

Education

PhD **Harvard University**, Astronomy & Astrophysics — Cambridge, MA
- Advisor: Charlie Conroy — Aug 2021 – May 2027 (exp.)

BS, BA **Johns Hopkins University**, Physics, Mathematics — Baltimore, MD
- Summa cum laude — Aug 2017 – May 2021
- Advisor: Kevin C. Schlaufman, Keith Hawkins

Research Interests

Stellar evolution and modeling, stellar ages and isochrone fitting, exoplanet ages, Galactic archaeology, asteroseismology, convection and mixing length theory, low mass stars, stellar spectroscopy, globular clusters

Teaching

- Harvard University — Lab Teaching Fellow (Spring 2024): ASTRON 16, Stellar and Planetary Astronomy
- Harvard University — Teaching Fellow (Spring 2022): ASTRON 100, Methods of Observational Astronomy

Honors and Awards

- Donald E. Kerr Memorial Award, Johns Hopkins University Department of Physics & Astronomy 2021
- Goldwater Scholar 2020

Talks

- Astro Coffee, Institute for Astronomy (IfA), University of Hawai‘i 2026
- Student Faculty Forum (StuFF), Center for Astrophysics | Harvard & Smithsonian 2026
- Exoplanets & Stars Seminar, Yale University 2026
- MMT Science Symposium, University of Arizona 2026
- MPIA Stars Meeting 2023
- ITC Luncheon Talk, Center for Astrophysics | Harvard & Smithsonian 2023
- Research Symposium, Maryland Space Grant Consortium 2018

Posters

- Searching for r-process Enhanced Metal-poor Stars in the Milky Way Halo, AAS 235 2020

Workshops & Conferences

- Yale/CeNAM Nuclear Asteroseismology Workshop, Yale University, New Haven, CT 2026
- MESA Summer School, KU Leuven, Belgium 2025
- Binary Stars in the Space Era, Keele University, UK 2025
- SDSS-V Collaboration Meeting, Flatiron Institute, NYC, NY 2023
- 235th AAS Meeting, Honolulu, HI 2020

Observatory Allocations

- Magellan Observatory, MIKE, 9 nights 2023

First-Author Publications

Classical Cases of Non-solar Mixing Length in Low Mass Stars 2026
Woody, Rebecca, Conroy, C., Cargile, P. A., Dotter, A.
arxiv.org/abs/2609.13384 (submitted to ApJ; arXiv:2609.13384)

The Rapid Formation of the Metal-poor Milky Way 2025
Woody, Rebecca, Conroy, C., Cargile, P., et al.
ui.adsabs.harvard.edu/abs/2025ApJ…978..152W/abstract (ApJ, 978, 152)

The Age–Metallicity–Specific Orbital Energy Relation for the Milky Way's Globular Cluster System Confirms the Importance of Accretion for Its Formation 2021
Woody, Rebecca, Schlaufman, K. C.
ui.adsabs.harvard.edu/abs/2021AJ….162…42W/abstract (AJ, 162, 42)

Co-Author Publications

An Ancient Brown Dwarf Transiting a Metal-poor Thick-disk Star 2026
Adams Redai, J., Chandra, V., Yee, S. W., …, ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2026ApJ..1002L..41A/abstract (ApJL, 1002, L41)

All-sky Kinematics of the Distant Halo: The Reflex Response to the LMC 2025
Chandra, V., Naidu, R. P., Conroy, C., …, ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2025ApJ…988..156C/abstract (ApJ, 988, 156)

A Detailed Chemical Study of the Extreme Velocity Stars in the Galaxy 2024
Nelson, T., Hawkins, K., Reggiani, H., …, ***Woody, Rebecca***
ui.adsabs.harvard.edu/abs/2024MNRAS.532.2875N/abstract (MNRAS, 532, 2875)

Distant Echoes of the Milky Way's Last Major Merger 2023
Chandra, V., Naidu, R. P., Conroy, C., …, ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2023ApJ…951...26C/abstract (ApJ, 951, 26)

Dwarf Galaxy Archaeology from Chemical Abundances and Star-formation Histories 2023
Johnson, J. W., Conroy, C., Johnson, B. D., …, ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2023MNRAS.526.5084J/abstract (MNRAS, 526, 5084)

A Ghost in Boötes: The Least-Luminous Disrupted Dwarf Galaxy 2022
Chandra, V., Conroy, C., Caldwell, N., …, ***Woody, Rebecca***
ui.adsabs.harvard.edu/abs/2022ApJ…940..127C/abstract (ApJ, 940, 127)

The Stellar Halo of the Galaxy is Tilted and Doubly Broken 2022
Han, J. J., Conroy, C., Johnson, B. D., …, ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2022AJ….164..249H/abstract (AJ, 164, 249)

A Tilt in the Dark Matter Halo of the Galaxy 2022
Han, J. J., Naidu, R. P., Conroy, C., …, ***Woody, Rebecca***
ui.adsabs.harvard.edu/abs/2022ApJ…934...14H/abstract (ApJ, 934, 14)

Birth of the Galactic Disk Revealed by the H3 Survey 2022
Conroy, C., Weinberg, D. H., Naidu, R. P., ..., ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2022arXiv220402989C/abstract (submitted to OJAp; arXiv:2204.02989)

Live Fast, Die α-Enhanced: The Mass–Metallicity–α Relation of the Milky Way's Disrupted Dwarf Galaxies 2022
Naidu, R. P., Conroy, C., Bonaca, A., ..., ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2022arXiv220409057N/abstract (submitted to ApJ; arXiv:2204.09057)